

Model Report — Core Task

Crop Disease Detection Model

A ConvNeXt-Tiny classifier trained across three combined leaf-image datasets, evaluated on an internal held-out split and on a genuine field-condition test set to measure real-world generalization.

	MODEL ARTIFACT
	agrismart_convnext_tiny_final.pth
	CLASSES
	38 crop-disease / healthy classes

	ARCHITECTURE
	ConvNeXt-Tiny, ImageNet-pretrained
	REPORT DATE
	15 September 2026

Result Summary

Macro-F1 is the primary ranking metric; it is reported on two evaluation sets with very different character.

MACRO F1 — INTERNAL HELD-OUT TEST (in-distribution, 7,131 images) 0.9809 Accuracy 0.9856 · Weighted F1 0.9856	MACRO F1 — PLANTDOC FIELD TEST (out-of-distribution, 236 images) 0.5675 Accuracy 0.6356 · Weighted F1 0.6419
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The ~42-point gap between these two numbers is the intended difficulty of the challenge: a model that memorises lab-condition leaf photos will not automatically perform well on messy, real-world field photography. Closing this gap — not raising the in-distribution number further — is where most remaining work should go.

Task

Multi-class image classification of crop leaf photographs into one of 38 classes (37 disease classes plus healthy states) across 13 crop species. Given a single leaf image, the model returns one predicted class and a confidence score, paired with basic precautionary guidance for the farmer.

Dataset & Split

Three public sources combined into one 38-class label space, deduplicated, then stratified 80 / 10 / 10.

Source	Role	Images	Link
Hassan Ikram MyDataset	Primary source for 35 of the 38 classes	63,549	https://www.kaggle.com/datasets/hassanikram/my-dataset?select=MyDataset
PlantVillage (color)	Only source for 3 classes — Blueberry, Raspberry, Soybean (healthy)	6,963	https://www.kaggle.com/datasets/abdallahalidev/plantvillage-dataset
PlantDoc — train split	Folded into training only, to add real-world images to existing classes	2,316	https://www.kaggle.com/datasets/abdulhasibuddin/plant-doc-dataset

Total collected: 72,828 images → after removing 0 corrupted files and 1,526 exact MD5-duplicate files → 71,302 images across all 38 classes.

Split	Images	Share
Train	57,041	80.00%
Validation	7,130	10.00%
Test (internal, stratified)	7,131	10.00%

A second, fully independent evaluation set — PlantDoc's own test split (236 images, 27 of 38 classes) — was held out of training entirely and used only for the field-condition check. Class imbalance in the training set (152 to 6,339 images per class) was addressed with a class-balanced WeightedRandomSampler using inverse-square-root weighting.

Model & Approach

Two-stage transfer learning on a ConvNeXt-Tiny backbone.

Architecture

- **ConvNeXt-Tiny**, ImageNet-pretrained (torchvision)
- **Classifier head replaced:** Linear(768 → 38)
- **27,849,350** total parameters
- **Input:** 224×224 RGB, ImageNet mean/std normalization

Augmentation (train only)

- **RandomResizedCrop**, horizontal flip
- **±15° rotation**, color jitter
- **Random erasing** ($p = 0.20$)

Stage 1 — head warm-up

- **Backbone frozen;** only the 29,222-parameter head trained
- **AdamW + cosine schedule**, 3 epochs
- **Best validation macro-F1:** 0.9488

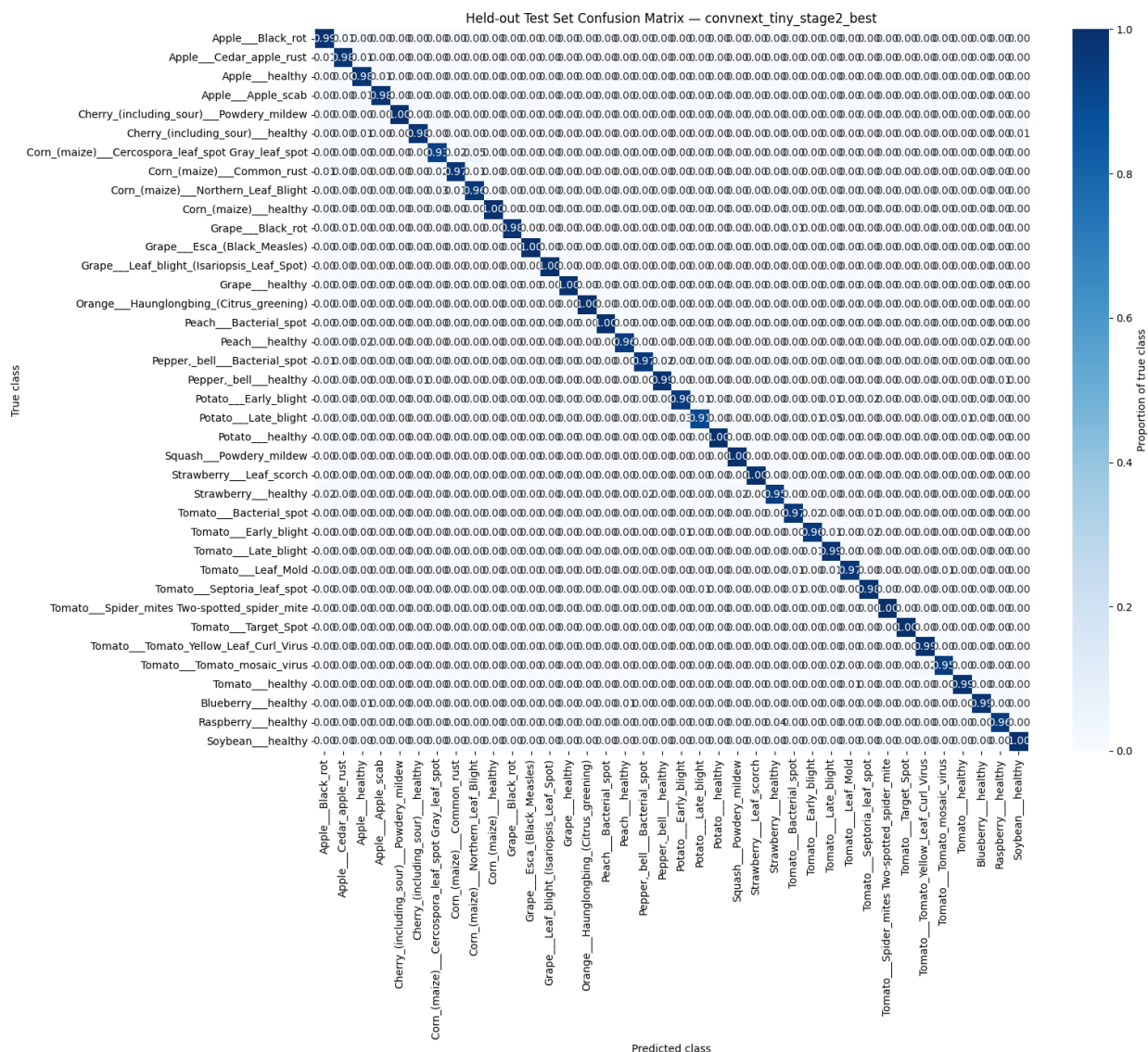
Stage 2 — full fine-tune

- **Entire network unfrozen**, differential learning rates
- **Early stopping** on validation macro-F1 (patience 3), 8 epochs run
- **Best validation macro-F1:** 0.9833

Loss: cross-entropy with label smoothing (0.1). Training used mixed precision (AMP), gradient clipping, and best-checkpoint selection by validation macro-F1.

Results — Internal Held-out Test

Stratified 10% split, in-distribution (same three source datasets as training).

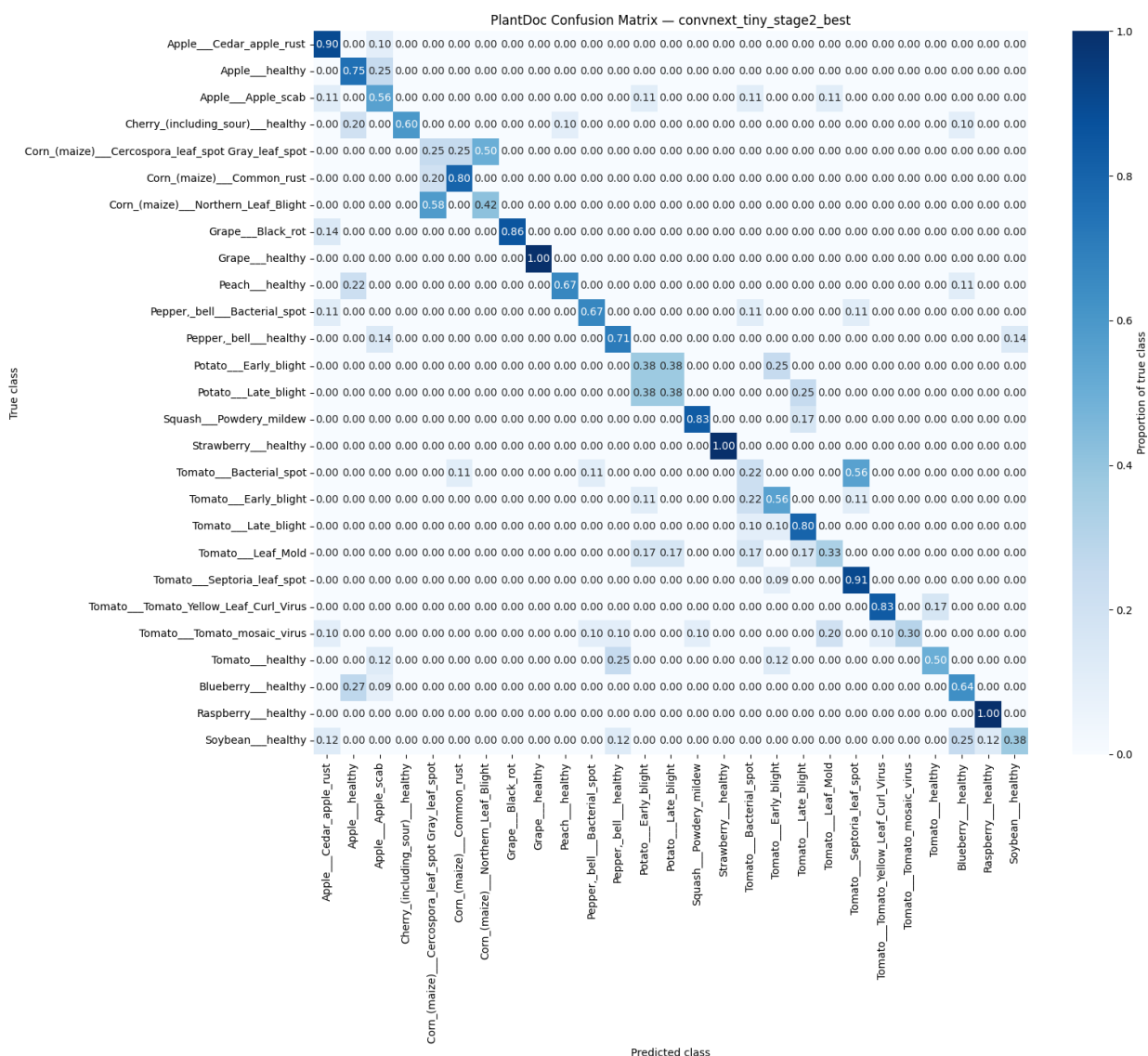


Confusion matrix, normalized by true class — internal held-out test set (7,131 images, 38 classes)

Per-class F1 ranges from 0.90–1.00 across all 38 classes. The weakest class is Corn (maize) — Cercospora leaf spot / Gray leaf spot (F1 = 0.92), which also has the smallest test support (58 images).

Results — Field Test (PlantDoc)

Out-of-distribution check: natural lighting, clutter and occlusion, never seen during training.



Confusion matrix, normalized by true class — PlantDoc held-out test split (236 images, 27 of 38 classes present)

Per-class F1 is far more variable here — from 1.00 (Grape — healthy, 12 samples) down to 0.24 (Tomato — Bacterial spot, 9 samples) — reflecting both the small sample size and the genuine difficulty of field photography.

Baseline & Limitations

Baseline. No organizer-published baseline macro-F1 was available at the time of writing. The two figures in this report (0.9809 in-distribution vs. 0.5675 out-of-distribution) serve as an internal baseline pair. Update this section once the organizers publish their private field-set baseline, and state where the submitted predict.py score falls relative to it.

Limitations

- **Lab-to-field generalization gap.** The ~42-point macro-F1 drop between the two test sets shows the model has not fully closed the gap the problem statement is built around; more field-style augmentation or field-condition training data would likely help.
- **Small, uneven field sample.** The PlantDoc check covers only 236 images and 27 of 38 classes, so per-class OOD numbers for classes with 4–10 samples are noisy.
- **Residual class imbalance.** Some classes (e.g. Potato — healthy, Grape — healthy) still have very low raw image counts even after weighted sampling.
- **Single-label predictions.** The model assumes one dominant label per image and does not detect multiple co-occurring symptoms or severity.
- **No calibration check.** Confidence scores are raw softmax outputs and should be read as relative, not calibrated probabilities.